

Explainable Artificial Intelligence for Interpreting Machine Learning Models

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Abstract—The recent proliferation of machine learning (ML) models in healthcare has been of great help in predictive performance on diagnostic and decision-support tasks. Nevertheless, most of the high-performing models are black-box systems, which restrict transparency and clinical trust. Interpretability is critical in safety-sensitive aspects like medicine to provide accountability and conformity to the existing medical knowledge. The proposal presented in this paper is to create a transparent machine learning system, which combines a Random Forest classifier and model-agnostic explainability methods. It is a binary classification task over a real-world medical dataset and interpretability is improved with SHapley Additive exPlanations (SHAP) to get global feature-importance and Local Interpretable Model-agnostic Explanations (LIME) to get instance-level interpretation. The experimental findings show that the developed framework is highly predictive (Accuracy: 0.964, F1-score: 0.963) but can give significant explanations to the prediction made by the model. The results indicate that the combination of ensemble learning and the use of post-hoc explanation methods can be used to balance predictive performance and transparency in healthcare use.

Index Terms—Explainable AI, Machine Learning, Interpretability, SHAP, LIME

I. INTRODUCTION

Machine learning (ML) is now an effective tool of unanswered difficulties in numerous domains, such as healthcare, finance, cybersecurity, and decision support systems [1]. ML models are capable of adapting to the trends of historical data and reaching a high degree of predictive accuracy, not only that, but also automatizing the processes previously performed by humans. Due to this, there is an increasing application of such models in practice where crucial decisions need to be correct and timely.

Machine learning models have been effective, however, they are not very transparent in most cases. Complex models such as ensemble methods and deep learning architectures are by definition black box and it is not always possible to interpret how the specific predictions are made by the user on the blackbox Problem. Such an inability to interpret is a severe problem that raises concerns regarding trust, accountability, fairness and ethical decision-making especially in sensitive fields like the sphere of healthcare.

Explanation of decision of a model in medicine is as important as in decision making. Clinicians and medical workers require interpretable information to validate predictions, study

of the factors that influence it and to ensure that the model functions according to known medical rules to ensure the predictive behavior is appropriate [2]. The low confidence levels in the output of highly accurate models prevent the use and rejection of these models in clinical settings because they cannot be explained.

The issue of interpretability in machine learning has led to explainable Artificial Intelligence (XAI) as a feasible solution. XAI methods aim to provide an explainable description of a model and behavior in an understandable format by revealing critical aspects of the model, detailing their reasoning, and describing their actions by humans [3]. Such explanations can be generated at a global level in regards to the overall behavior of the model, locally in regards to specific predictions. That is why the model-agnostic versions of the XAI have become particularly helpful due to the possibility to use them with the vast majority of the machine learning algorithms that do not require any modification of the underlying model whatsoever. [4], [5]

The given paper presents a transparent AI scheme to decode the machine learning models using the model-agnostic explanation techniques. A binary classification is performed using a Random forest classifier that is trained on a real world medical data. To enhance the transparency, SHapley Additive exPlanations (SHAP) are used to rank the global feature, and the Local Interpretable Model-agnostic Explanations (LIME) are used to interpret the individual predictions. The recent studies have also helped to highlight the necessity of ethical AI implementation in the safety-sensitive sectors. Transparency and explainability have become one of the principles of design, not a feature, and required by regulatory and ethical considerations [6], [7]. Specifically, machine learning systems with a healthcare focus should be able to make predictive results and results clear and consistent with established medical knowledge in a clinical way [8]. Such increased pressure on reliable AI systems and systems encourages the adoption of explainability methods into predictive modeling pipelines.

II. LITERATURE STUDY

Machine learning has become fast adopted in key areas like healthcare, finance and decision support systems [1], [9]. Most

of the advanced models such as ensemble and deep neural networks are black-box systems and offer limited interpretability and trust even though they perform better on predictions than others [7], [10]. This vagueness is particularly objectionable in the safety-related contexts where responsibility and equality are crucial.

Rudin argued that in situations where there are inherently interpretable alternatives, black-box models should not be used in making high-stakes decisions [10]. As another important point, Doshi-Velez and Kim [11] stressed that a strict scientific framework is required to measure interpretability in machine learning systems.

There are a number of surveys that have systematically enumerated explainable AI (XAI) techniques. Guidotti et al. offered a taxonomy of intrinsic and post-hoc explanations. The classification was extended by Linardatos et al. who distinguished between global and local interpretability methods [3]. Arrieta et al. [7] have provided a detailed review of challenges, opportunities and open research directions in XAI.

Interpretability has become a key element in creating clinical trust in the healthcare field. Caruana et al. [12] have shown that interpretable models are capable of achieving competitive performance in medical risks prediction problems. The part by Holzinger et al. emphasized the significance of medical decision support systems which can be explained, with transparency being a pre-condition to clinical acceptance [8].

LIME and SHAP are some of the most commonly used post-hoc explanation methods, including LIME and SHAP. LIME offers local surrogate explanations of single predictions, whereas SHAP uses the cooperative game theory to provide consistent feature importance values. Empirical research has demonstrated SHAP to meet desirable theoretical characteristics (local accuracy and consistency) [5].

The algorithm of Random Forest developed by Breiman (2001) has also been identified as having a good predictive potential and with built-in measures of importance of features. Nevertheless, the ensemble methods can also be supplemented with other explainability methods to make it more transparent in sensitive areas.

Ethical issues and regulatory concerns of AI systems have also been discussed in recent literature, but with a focus on fairness, accountability, and transparency as the key conditions of the use of AI systems in the modern world [2]. Besides post-hoc explainability methods, researchers have investigated inherently interpretable models as an alternative to complicated black-box systems as well [10]. Nevertheless the empirical evidence shows that ensemble techniques like the Random Forests tend to perform well in structured medical data, with better predictive accuracy in general [13]. As a result, in recent years, hybrid models that integrate high-performing predictive models with model-agnostic explanation methods have become of particular interest to researchers [14]. The frameworks are designed to offer a trade-off between predictive accuracy and transparency, especially in the context of healthcare where both efficiency and transparency seem to be imperative.

However, in spite of these tremendous developments, there

is still the necessity of practical models that can combine predictive performance with interpretable explanations in practical healthcare applications. This research paper is an advancement of the previous literature, as it integrates ensemble learning with model-agnostic explanation methods to trade off between performance and interpretability.

A. Framework Overview

The general process of the proposed XAI is shown in Fig. 1. The model is executed in a consecutive pipeline whereby the input data undergoes a preprocessing stage before being utilized to train a machine learning model. After model prediction, explainability methods are used to give an insight on the decision-making process.

B. Steps Involved

1) Data Loading: In this study, a publicly available medical dataset is utilized. The data is represented by several numerical variables that present the characteristics of clinical features, and a dichotomous predictive variable that indicates the diagnostic outcome. Data is loaded and checked to make sure that it is consistent and correct.

2) Data Preprocessing: The dataset is subjected to preprocessing measures such as irrelevant attributes elimination, missing values, and feature scaling before it is model-trained. Normalization of feature values is done using standardization that assists in enhancing the stability and performance of models.

3) Model Training: The training of a Random Forest classifier is done with the processed dataset. The data is split into training and testing data sets and most data is used during the training and the rest of it is allocated during the evaluation.

4) Prediction Generation: This phase involves generating predictions about the possible futures. This stage entails making predictions of the potential futures. Training leads to model predictions on the test data. Accuracy, precision, recall, and F1-score are some performance metrics that are calculated to determine classification effectiveness.

5) Global Explainability: Description: The global explainability with SHAP is given in this paper. SHapley Additive exPlanations (SHAP) are used to examine the importance of global features. SHAP values help to understand the importance of each feature to the model predictions which help to get a general idea about what features have the greatest impact on decision-making.

6) Local Explainability Using LIME: The linear model is more likely to predict price changes than the nonlinear model. Local explainability through LIME: The linear model will tend to forecast changes in prices than the nonlinear model. Local Interpretable Model-agnostic Explanations (LIME) are used to elucidate single predictions. The input produced by a local model is interpreted by LIME across the specific instances and therefore users can gain an insight into why the local model made a certain prediction based on a given input.

The proposed framework provides an opportunity to make model decisions that will be accurate and interpretable by combining predictive modeling with explainability methods. This will contribute to informed decision making in healthcare-oriented machine learning applications and transparency. In the recent past, explainable artificial intelligence has seen new developments, highlighting how to incorporate interpretability into the predictive modeling processes. Instead of applying explainability as a post-processing solution, current studies promote systematized analysis of transparency, as well as accuracy indicators. These strategies are especially applicable to healthcare-related uses, where interpretability is a direct determinant of practitioner trust and adoption of the system. .

III. METHODOLOGY

It is described here as what data was used to conduct the study, the experimental setup, and the metrics associated with the evaluation of the performance of the model. All were done in Python based machine learning libraries.

A. Dataset Description

The proposed explainable machine learning framework was tested using the Breast Cancer Wisconsin Diagnostic (WDBC) data. The sample size is 569, which is a collection of digitized images of fine needle aspirate (FNA) biopsies of breast masses. The samples are examples of tumors that are characterized by quantitative morphological features.

The samples include 30 numerical real valued features that are based on cell nuclei properties including radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension. Each characteristic is furnished with three measurements which is the mean score, standard error and worst score thus forming a full set of features that can be used in classifications and interpretability analysis.

The target variable is a binary response with malignant tumors being classified under class 1 and benign tumors under class 0. Among all the samples, there are 357 benign cases and 212 malignant cases, which provide a moderate level of class imbalance. This imbalance underlines the value of other metrics of evaluation than accuracy, including precision, recall, and F1-score.

There are no missing values in the dataset, which means that the data is consistent, and less time is spent on preprocessing. Thanks to its medical applicability, its clean format, and its broad use in research, the WDBC dataset would be suitable to both predictive performance and explainability of machine learning models.

IV. DATA PREPROCESSING

Before the model has been trained, a number of preprocessing were conducted to achieve data quality and consistency. The attribute of unique identifier was dropped because it is not a predictive attribute. Diagnostic labels were coded into numerical values whereby, malignant samples were coded 1 and benign samples coded 0.

Each of the feature variables was numerical and hence not to be categorically coded. To normalise the range of features and enhance stability of the models, standardization was used to scale features. The use of standardization means that attributes that have a higher numerical range do not prevail in the learning process.

Lastly, a stratified train-test split was used to split the dataset so that the proportion of the classes was held constant. The above preprocessing steps make the machine learning model more robust and generalized.

A. Train-Test Split

The dataset was split in terms of an 80:20 ratio to form a training and testing set. The machine learning model was constructed using the training set and the performance was evaluated using the testing set. This division has been done to make sure that the generalization ability of the model is measured on unseen data. In the split, stratified sampling has been used to maintain the initial class distribution in training and testing sets. In this way, the results of evaluations will still reflect the actual diagnostic conditions in the real world and will not be biased towards the majority group.

B. Evaluation Metrics

Some standard measures were employed to assess the performance of the classification model, among them being accuracy, precision, recall, and F1-score. These measures give a holistic view of prediction ability of the model particularly when used in a medical classification where falsely predicting can result in serious consequences. Accuracy, in a medical diagnosis task, can be misleading because of a class imbalance. Thus, accuracy and recall are essential measures that can be used to assess the accuracy of prediction, especially of malignant cases. The F1-score offers a more balanced evaluation through a combination of false positives and false negatives, so it is quite suitable to use in the healthcare context.

V. MODEL ARCHITECTURE AND HYPERPARAMETER SETTINGS

The primary prediction model of the selected type was a Random Forest classifier because it is well-performing and can address non-linear relationships, not to mention that it performs well with tabular data. Random Forest is an ensemble learning algorithm that builds numerous decision trees and combines their forecasts to achieve less variance and better generalization.

In this paper model was set up with 100 decision trees. Split quality in each tree was measured using the Gini impurity measure. To learn intricate patterns existing in the data, trees were left to be grown without depth limitation. A random seed was fixed to be able to replicate the results.

Random Forest features importance scores of the features in the ensemble, which means without any additional post-hoc explainability methods. The following features make the Random Forest a good option in the context of both accurate prediction and interpretability in healthcare applications.

VI. EXPLAINABILITY TECHNIQUES

The goal of explainable Artificial Intelligence (XAI) methods is to give insight into machine learning models by describing the process of producing predictions. Model-agnostic explanation methods were used in this study to guarantee that the techniques can be used with any classifier and to also be flexible.

A. Global Explainability Using SHAP

A popular explainability approach is SHapley Additive exPlanations (SHAP), which is a cooperative game theory approach. SHAP attaches to every feature a contribution value which is a measure of its influence on the prediction of the model. The aggregate of SHAP values in instances can be used to get global feature importance.

SHAP meets some important theoretical properties like consistency and local accuracy, which make it a reputable method of interpreting complex models. SHAP summary plots were utilized in this research to determine the most powerful clinical features that yielded to diagnosis results.

B. Local Explainability Using LIME

Local Interpretable Model-agnostic Explanations (LIME) is the focus on explaining individual predictions by localizing the model around the surrogate that can be understood. LIME perturbation values features and monitors prediction changes to determine which features are influential to an instance.

Local explanations are especially useful at healthcare centers, where it is necessary to justify patients personally. With the application of LIME, clinicians can know the motivation behind a specific diagnostic choice that is applied to a given case.

VII. RESULTS

The performance of the proposed explainable machine learning framework was evaluated on the test dataset.

A. Performance Evaluation

The performance of the Random Forest classifier was evaluated using standard classification metrics. Table I summarizes the evaluation results obtained on the test dataset.

TABLE I
PERFORMANCE METRICS FOR RANDOM FOREST MODEL

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	0.964	0.972	0.955	0.963

Fig. 1 illustrates the confusion matrix of the Random Forest classifier, indicating a high number of correct predictions for both classes. Fig. 2 presents the SHAP summary plot, highlighting the most influential features contributing to model decisions. Fig. 3 shows a local explanation generated using LIME, providing insight into an individual prediction made by the model.

The Random Forest classifier had high accuracy in classification and this indicates that the classifier is effective in predicting results of diagnosis.

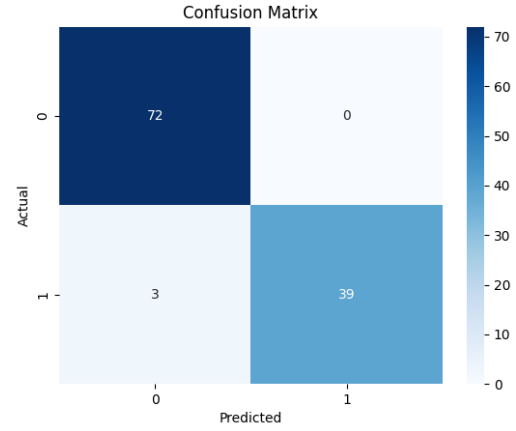


Fig. 1. Confusion Matrix for Random Forest Classifier

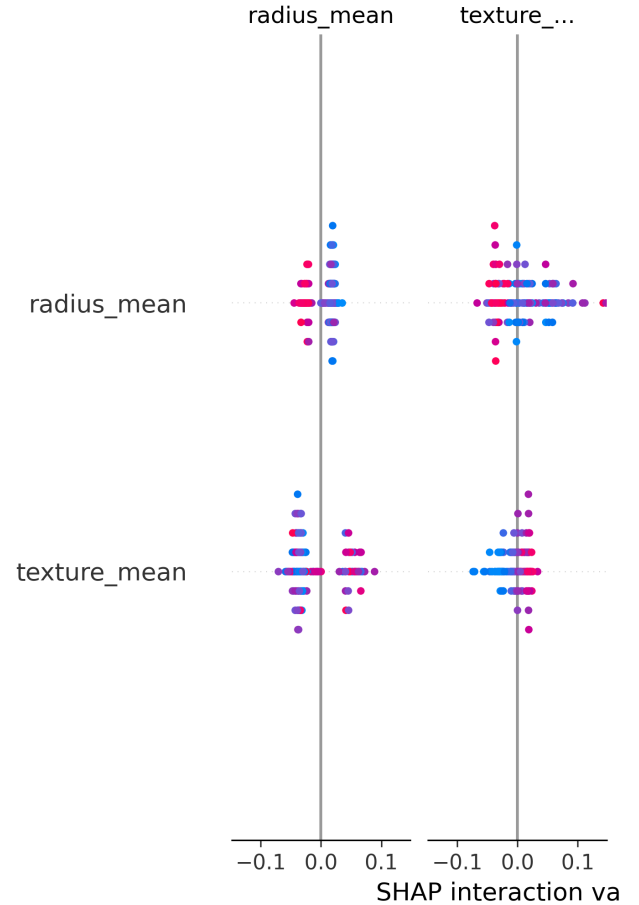


Fig. 2. SHAP Summary Plot Showing Global Feature Importance

Along with the results of predictive performance, SHAP and LIME methods were used to analyze explainability. Global explanations using SHAP showed the most significant features that explained the predictions of the model. Aspects of cell radius, texture and concavity were found to play a major role in classification decisions.

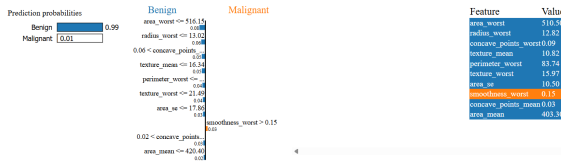


Fig. 3. LIME Explanation for an Individual Prediction

Local explanations with the help of LIME were able to give instance-level information on single predictions. These descriptions were used to demonstrate the way in which certain feature values would be used in determining the decision of the model on a particular sample. The mixed global and local explanations made the model more transparent and understandable regarding the machine learning model.

The findings show the suggested framework is effective in balancing predictive accuracy and explainability, which is why it can be used in healthcare decision-support systems.

VIII. COMPARATIVE ANALYSIS

An additional evaluation of the effectiveness of the proposed approach was carried out through a comparative analysis with the help of Logistic Regression as a control model. Logistic Regression is inherently interpretable and has a weakness in non-linear relationships.

It has been found experimentally that though the performance of Logistic Regression can be reasonable, it is not better than the performance of the Random Forest classifier regarding accuracy and F1-score. Random Forest model has been proved to be more competent in learning complex interactions of features.

Through a combination of the predictive power of Random Forest and post-hoc explanatory methods, the proposed framework balances the performance and interpretability models and is, therefore, appropriate to real-world medical decision-support systems.

IX. DISCUSSION

These experimental findings indicate that machine learning models can attain high predictive accuracy and at the same time be explainable by utilizing explainable AI methods. Random Forest classifier was a good choice in this study because it is strong and has the capacity to work with tabular medical data.

SHAP acquired global explanations were informative about the general actions of the model and made it possible to find some important characteristics affecting the predictions. The information would help healthcare practitioners to know what drives the results of diagnostic processes. Explanations created by LIME locally also promoted transparency because they further clarified single individual predictions, which is especially vital in patient-specific decision-making.

The framework proposed can solve the trust-related issues with black-box models by incorporating explainability into the

machine learning pipeline. The findings can indicate that explainable AI may help fill the divide between model accuracy and human trust and lead to the application of AI in real-life conditions in areas of sensitive applications. The reason is that the integration of techniques of explainable AI can dramatically increase the transparency of the suggested framework. The model helps to make informed clinical decisions by specifying the influential features and offering instance-level explanations. As the results are encouraging, the future studies should perform additional validation on more substantial and more varied datasets to assess the generalizability.

X. ETHICAL CONSIDERATIONS AND PRACTICAL IMPLICATIONS

The application of machine learning in healthcare creates ethical issues connected to trust, responsibility and justice. Evaluation of incorrect or biased predictions can result in poor clinical outcomes, and that is why transparent and interpretable models are important.

Explainable AI methods can also be used to reduce these fears by allowing users to make informed decisions about the model. The proposed framework will lead to responsible AI usage and facilitate ethical adoption of AI in clinical settings because it offers both global and local explanations.

XI. CONCLUSION

This paper introduced an explainable artificial intelligence framework for interpreting machine learning predictions in healthcare applications. A medical dataset was used to train a Random Forest classifier to be used with binomial classification, and model-agnostic explainability methods were used to improve the transparency.

SHAP allowed the interpretation of feature importance globally, whereas LIME allowed one to get the local explanations of individual predictions. Experimental findings revealed that the proposed method is highly classified, and has interpretive and comprehensible explanations. As the findings suggest, explaining the work of machine learning systems is vital in enhancing trust, reliability, and useful practical application in the healthcare field.

XII. FUTURE SCOPE

Subsequent endeavors can be employed to expand this study to explainable AI methods of deep learning models and more convoluted data. Other XAI tools like counterfactual explanations, rules can be considered to further promote interpretability. The suggested framework would be applicable to real-time clinical decision-support systems, which would allow ongoing monitoring and clarification of the predictions. Incorporation of human medical knowledge by the professionals might contribute even more to the efficacy and adoption of explainable machine learning models in practice.

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